Quantum Game Theory

Equilibrium Search, Optimization, and Learning

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Introduction

Exploration

The goal is to analyze equilibria in classical and quantum strategic games by investigating these theoretical questions:

1. Nash equilibrium after introducing entanglement
2. Eisert–Wilkens–Lewenstein (EWL) protocol
3. Equilibrium search: learning or optimization problem?

Background

Classical prisoner's dilemma

Table 1: Possible scenarios from actions {Cooperate, Defect}.

	Decision	Outcome
1.	Both cooperate	**Reward:** both banned for 1 month
2.	One defects, one cooperates	**Temptation:** defector receives warning; **Sucker's payoff:** cooperator permabanned
3.	Both defect	**Punishment:** both banned for 1 year

Classical prisoner's dilemma (cont.)

Ban duration is inversely related to payoff. Define

$$T = 3, \qquad R = 2, \qquad P = 1, \qquad S = 0,$$

so that

$$T > R > P > S.$$

Bonnie \ Clyde	Clyde cooperates	Clyde defects
Bonnie cooperates	$R, 2$ (Bonnie); $R, 2$ (Clyde)	$S, 0$ (Bonnie); $T, 3$ (Clyde)
Bonnie defects	$T, 3$ (Bonnie); $S, 0$ (Clyde)	$P, 1$ (Bonnie); $P, 1$ (Clyde)

Figure 1: Payoff matrix for Bonnie and Clyde.

Non-Zero Sum Symmetric Games

Unlike prisoner's dilemma, a game of chicken contains multiple Nash equilibria and strategic coordination behavior.

P_1 \ P_2	P_2 swerves	P_2 straight
P_1 swerves	0 (P1), 0 (P2)	-1 (P1), 1 (P2)
P_1 straight	1 (P1), -1 (P2)	-1000 (P1), -1000 (P2)

Figure 2: Payoff matrix for the game of chicken played by players P_1 and P_2.

Metrics

A strategy profile s is a **Nash equilibrium** if no player can improve by changing strategy alone:

$$u_i(s_i, s_{-i}) \geq u_i(s'_i, s_{-i}).$$

A strategy profile is **Pareto efficient** if no other outcome makes one player better off without making another worse off.
In prisoner's dilemma:

$$(D, D) \text{ is Nash equilibrium,}$$

but

$$(C, C) \text{ is Pareto superior.}$$

This paradox is known as the prisoner's dilemma.

Quantum Computing

Quantum computing processes information using principles from quantum mechanics or quantum phenomena, especially:

- **Superposition**
- **Entanglement**
- **Measurement**

Superposition: Bits vs. Qubits

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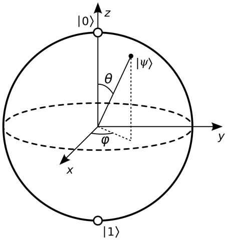
0

1

Classical bit stores either 0 or 1



Bloch sphere of a qubit

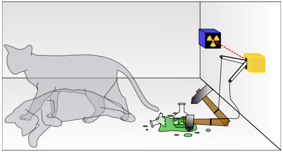
A qubit may exist in a superposition:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1.$$

Measurement Problem

A quantum state may contain multiple possible outcomes simultaneously until measured, collapsing the state probabilistically:

$$|0\rangle \text{ with probability } |\alpha|^2, \qquad |1\rangle \text{ with probability } |\beta|^2.$$



Schrödinger's cat: superposition before observation

Interpretation

Quantum systems evolve continuously until measurement.

Entanglement

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Two quantum systems are **entangled** when their joint state cannot be separated into independent components:

$$|\psi\rangle \neq |\psi_A\rangle \otimes |\psi_B\rangle.$$

This means:

- measuring one system reveals information about the other,
- correlations exist beyond classical independence,
- strategies may become correlated through entanglement.

Connection to Quantum Game Theory

Entanglement changes the strategic correlation structure of a game and may alter equilibrium behavior.

Quantum Game Theory

From Classical to Quantum

Classically, a player chooses:

$$C \quad \text{or} \quad D.$$

In the quantum version, the basis states represent classical actions:

$$|C\rangle = \text{cooperate}, \qquad |D\rangle = \text{defect}.$$

A general quantum state may be a superposition:

$$|\psi\rangle = \alpha|C\rangle + \beta|D\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1.$$

Players do not directly choose outcomes. They apply unitary operations to qubits, and outcomes appear only after measurement.

EWL Quantum prisoner's dilemma

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Eisert, Wilkens, and Lewenstein proposed the following protocol:

1. Start with the classical state:

$$|CC\rangle.$$

2. Apply an entangling operator J.
3. Each player applies a local unitary strategy:

$$U_A \otimes U_B.$$

4. Undo the entanglement with $J^\dagger$, then measure.

The final state is:

$$|\psi_f\rangle = J^\dagger (U_A \otimes U_B) J |CC\rangle.$$

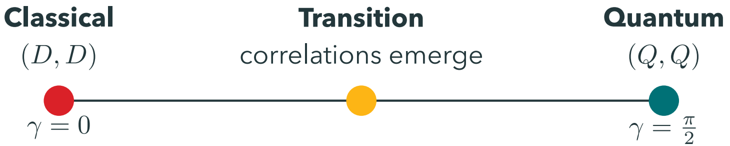
Entanglement Parameter

Let D denote the defection operator. The entangling operator is commonly written as:

$$J(\gamma) = \exp\left(-i\frac{\gamma}{2} D \otimes D\right), \qquad \gamma \in \left[0, \frac{\pi}{2}\right].$$

$$\begin{aligned} \gamma &= 0 && \Rightarrow \quad \text{classical game,} \\ \gamma &= \frac{\pi}{2} && \Rightarrow \quad \text{maximally entangled quantum game.} \end{aligned}$$

Entanglement Visualized



Interpretation

Increasing entanglement changes the strategic correlation structure of the game, shifting equilibrium behavior from classical defection toward cooperative quantum equilibria under restricted EWL strategy.

Quantum Strategies

Players choose unitary transformations of the form:

$$U(\theta, \phi) = \begin{pmatrix} e^{i\phi}\cos(\theta/2) & \sin(\theta/2) \\ -\sin(\theta/2) & e^{-i\phi}\cos(\theta/2) \end{pmatrix}.$$

Classical cooperation and defection are special cases:

$$C = I, \qquad D = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

EWL also identifies a quantum strategy:

$$Q = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}.$$

Joint Quantum Strategy Space

Players act on qubits in the joint Hilbert space:

$$\{|CC\rangle, |CD\rangle, |DC\rangle, |DD\rangle\}.$$

	$|C\rangle_{P_2}$	$|D\rangle_{P_2}$
$|C\rangle_{P_1}$	$|CC\rangle$	$|CD\rangle$
$|D\rangle_{P_1}$	$|DC\rangle$	$|DD\rangle$

Measurement collapses the quantum state into **one** classical outcome.

Expected Payoffs

After measurement, the final quantum state produces probabilities for

$$P_{CC}, \qquad P_{CD}, \qquad P_{DC}, \qquad P_{DD}.$$

Player A's expected payoff is:

$$\mathbb{E}[u_A] = RP_{CC} + SP_{CD} + TP_{DC} + PP_{DD}.$$

Player B's expected payoff is:

$$\mathbb{E}[u_B] = RP_{CC} + TP_{CD} + SP_{DC} + PP_{DD}.$$

Equilibrium Search as Optimization

For fixed opponent strategy U_B, player A solves

$$U_A^* \in \arg\max_{U_A \in \mathcal{S}} \mathbb{E}[u_A(U_A, U_B; \gamma)].$$

A Nash equilibrium is a pair (U_A^*, U_B^*) such that

$$\mathbb{E}[u_A(U_A^*, U_B^*)] \geq \mathbb{E}[u_A(U_A, U_B^*)],$$

$$\mathbb{E}[u_B(U_A^*, U_B^*)] \geq \mathbb{E}[u_B(U_A^*, U_B)].$$

Interpretation

Since the admissible strategy space is continuous, equilibrium search becomes a nonconvex policy optimization problem rather than finite action selection.

Numerical Experiment

To make the theory testable, the project can simulate the EWL game over a grid of strategy parameters:

1. Sweep entanglement values $\gamma \in [0, \frac{\pi}{2}]$.
2. Grid-search player strategies $U(\theta, \phi)$.
3. Compute $|\psi_f\rangle = J^\dagger (U_A \otimes U_B) J |CC\rangle$.
4. Convert amplitudes into outcome probabilities:

$$P_{CC}, P_{CD}, P_{DC}, P_{DD}.$$

5. Plot expected payoffs and approximate best responses.

Expected Output

Increasing γ changes best-response behavior and when (Q, Q) becomes stable under the restricted EWL strategy set.

Literature Review and Analysis

Paper 1: Meyer, *Quantum Strategies*

Meyer, one of the earliest pioneers of quantum advantage in game theory, proposed that a player with access to quantum strategies can outperform an opponent restricted to classical strategies.

Interpretation

His work motivated the broader idea that quantum mechanics can fundamentally expand strategic action spaces, raising the question: what happens when **all** players have quantum strategies?

Paper 2: Eisert–Wilkens–Lewenstein

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EWL introduced a quantum version of prisoner's dilemma. Under *maximal* entanglement and a restricted strategy space, the classical equilibrium (D, D) can be replaced by the quantum equilibrium (Q, Q), at which both players receive the cooperative payoff (R, R).

Interpretation

Entanglement allows stable cooperation *without* communication.

Paper 3: Benjamin–Hayden Critique

Benjamin and Hayden argued that the EWL result depends on the restricted allowable strategy space. If players are permitted to use a larger class of unitary operations, then:

- the proposed quantum equilibrium may no longer be stable,
- counter-strategies may emerge,
- cooperative outcomes become less automatic.

Interpretation

This critique reframes the project from "quantum games solve prisoner's dilemma" into an analysis of underlying assumptions, admissible strategy spaces, and equilibrium stability.

Restricted vs. Expanded Strategy Spaces

Restricted EWL Space

- Strategies use the two-parameter family $U(\theta, \phi)$
- Classical C, D and quantum Q are included
- (Q, Q) can become stable at maximal entanglement

Expanded Unitary Space

- Players may access a larger class of local unitaries
- Counter-strategies can appear
- Cooperative equilibrium is no longer automatic

Numerical Implication

Results should explicitly state which strategy space was searched; otherwise, claims about equilibrium stability become ambiguous.

Research Direction

Learning Dynamics

Parameterize each player's strategy by (θ, ϕ) and update parameters using payoff feedback:

$$(\theta_A, \phi_A) \leftarrow (\theta_A, \phi_A) + \eta \nabla_{(\theta_A, \phi_A)} \mathbb{E}[u_A].$$

$$(\theta_B, \phi_B) \leftarrow (\theta_B, \phi_B) + \eta \nabla_{(\theta_B, \phi_B)} \mathbb{E}[u_B].$$

Connection to Learning

This resembles conceptual policy-gradient dynamics in multi-agent learning: agents iteratively update parameterized strategies using observed payoff signals.

Deliverables

Worst-case: rigorous theoretical comparison and literature review.

Best-case: payoffs and learning trajectories from equilibrium search.

Learning Theory

Searching for a quantum strategy is solving an **optimization** problem:

$$\underset{U_A}{\operatorname{argmax}} \ \mathbb{E}[u_A(U_A, U_B)].$$

In **repeated** play, agents could update strategies similar to:

- reinforcement learning,
- policy optimization,
- gradient-based learning,
- multi-agent learning.

Interpretation

Quantum equilibrium search may be interpreted as learning over continuous unitary strategy spaces.

Classical vs. Quantum Strategies

A classical mixed strategy randomizes over actions:

$$pC + (1 - p)D.$$

A quantum strategy transforms amplitudes:

$$|\psi\rangle = \alpha|C\rangle + \beta|D\rangle.$$

The difference is that amplitudes can interfere before measurement.

Theoretical Claim

Unlike classical randomization, entanglement and interference can produce correlations unavailable to independent mixed strategies.

Takeaways

Limitations

- **Restricted strategy spaces:** EWL's cooperative equilibrium depends on which unitary strategies are allowed.
- **Curse of physicality:** Current quantum systems are affected by noise and decoherence.
- **Equilibrium instability:** Enlarging the strategy set may introduce counter-strategies.
- **Learning dynamics:** It remains unclear whether adaptive agents converge to stable quantum equilibria.
- **Interpretation:** Quantum advantage may depend on whether entanglement is treated as a resource provided before the game.

Open Problem

When does quantum game theory produce new strategic behavior rather than a reformulation of correlated/mixed classical strategies?

Conclusion

The EWL protocol, one of the foundational models in quantum game theory, suggests that entanglement can shift the classical equilibrium of prisoner's dilemma toward a cooperative quantum equilibrium.

However, the Benjamin–Hayden critique shows that this conclusion is not unconditional. The stability of the result depends critically on how the quantum strategy space is defined.

Main Takeaway

Quantum game theory may serve as a theoretical bridge between equilibrium analysis, optimization, and learning.

Future Work

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Although primarily theoretical, I aim to connect quantum game theory to reinforcement learning and computational optimization through neural policies over continuous unitary strategy spaces.

Next Steps

- Implement payoff-landscape visualizations over (θ, ϕ, γ).
- Analyze *how* entanglement changes best-response behavior.
- Study convergence and stability under policy-gradient updates.
- Test (Q, Q) stability under the restricted EWL strategy space.

References

[1] D. Meyer. *Quantum Strategies*. Phys. Rev. Lett., 82(5):1052–1055, 1999.
[2] J. Eisert, M. Wilkens, and M. Lewenstein. *Quantum Games and Quantum Strategies*. Phys. Rev. Lett., 83(15):3077–3080, 1999.
[3] S. Benjamin and P. Hayden. *Comment on "Quantum Games and Quantum Strategies"*. Phys. Rev. Lett., 87(6), 2001.
[4] J. Nash. *Non-Cooperative Games*. Ann. Math., 54(2):286–295, 1951.
[5] R. Sutton and A. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2018.
[6] M. Nielsen and I. Chuang. *Quantum Computation and Quantum Information*. Cambridge Univ. Press, 2010.
[7] L. Busoniu, R. Babuska, and B. De Schutter. *A Comprehensive Survey of Multiagent Reinforcement Learning*. IEEE Trans. Syst., Man, Cybern. C, 38(2):156–172, 2008.
[8] R. Jankowski. *Punishment in Iterated Chicken and Prisoner's Dilemma Games*. Rationality and Society, 2(4):449–470, 1990.
[9] S.-H. Lau and V.-L. Mui. *Using Turn Taking to Achieve Intertemporal Cooperation in Infinitely Repeated* 2×2 *Games*. Theory and Decision, 72(2):167–188, 2012.
[10] K. Ikeda and S. Aoki. *Infinitely Repeated Quantum Games and Strategic Efficiency*. Quantum Information Processing, 20(12), 2021.
[11] T. Ferguson. *A Course in Game Theory*. World Scientific, 2020.

Thank You!